



RKDF UNIVERSITY, RANCHI

New Scheme of Examination as per AICTE Flexible Curricula

MCA(Master in Computer Application)

SCHEME

First Semester

S.No.	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
			Max	Max	Min	Max	Min
1	MCA101	Discrete Mathematics	30	70	21	100	35
2	MCA102	Programming in C & C++	30	70	21	100	35
3	MCA103	Operating System	30	70	21	100	35
4	MCA104	Computer Architecture	30	70	21	100	35
5	MCA105	Accounting and Financial Management	30	70	21	100	35
Practical							
Total							
			Max		Min		
1	MCA151	Office management Lab	50		25		
2	MCA152	C & C++ Lab	50		25		
3	MCA153	Communication Skills	50		25		



RKDF UNIVERSITY, RANCHI

New Scheme of Examination as per AICTE Flexible Curricula

MCA(Master in Computer Application)

First Semester

Choice Based Credit System

MCA Year 1 Semester 1						
S.No.	Subject Code	SUBJECT NAME	PERIODS			Credit
			L	T	P	
1	MCA101	Discrete Mathematics	3	1	0	4
2	MCA102	Programming in C & C++	3	1	0	4
3	MCA103	Operating System	3	1	0	4
4	MCA104	Computer Architecture	3	1	0	4
5	MCA105	Accounting and Financial Management	3	1	0	4
Practicals						
1	MCA151	Office management Lab			2	1
2	MCA152	C & C++ Lab			2	1
3	MCA153	Communication Skills			2	1
Total					8	23



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

First Semester

Branch	Subject Title	Subject Code
MCA	Discrete Mathematics	MCA101

UNIT - I

Introduction to Discrete Mathematical Structures: Sets, Relations and functions- Sets, Types of Sets, Multisets, Operations on Sets, Relations and Properties of Relations, Representation of Relations, Equivalence Relation, Closures of Relations, Functions ,injection, Surjection and Bijective mapping, Composition of functions, Introduction to **characteristic functions.**

UNIT - II

Methods of Proof: Direct Proofs, Indirect Proofs, Mathematical Induction, Method of Contradiction. Permutations and Combinations, Pigeon Hole Principle, Principle of Inclusion and Exclusion, Sequence and Series, Generating Functions.

UNIT - III

Mathematical Logic Proposition and Propositional Calculus: Posets and Lattices: Partial Order Set, Bounding Elements, Well Ordered Set, Topological Sorting, Lattices, Principle of Duality, Bounded, Distributed, and Complemented Lattices

UNIT - IV

Graph Theory: Types of Graphs, Path and Circuits, Eulerian Path and Circuits, Hamiltonian Path and Circuits, Shortest Path Algorithms

UNIT - V

Group: Definitions and Properties, Coset & Subgroup, Normal subgroup, Homomorphism of groups, Cyclic Group, Permutation Group. Matrix Algebra: Matrices,

Rank of Matrix, Solving System of Equations-Eigen Values and Eigenvectors-Inverse of a Matrix - Cayley Hamilton Theorem

Text Book:

1. Keneth H. Rosen, "Discrete Mathematics and Its Applications", TMH, 2012

References:

1. C L Liu, "Elements of Discrete Mathematics", TMH, 2012.
2. Kolman, Busby & Ross, "Discrete Mathematical Structures", PHI, 2015.
3. Narsingh Deo, "Graph Theory With Application to Engineering and Computer Science", PHI, 1979



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

First Semester

Branch	Subject Title	Subject Code
MCA	Programming in C & C++	MCA102

UNIT - I

Programming fundamentals through C: Algorithms and Flowcharts, Data types, constants, variables, operators, data input and output, assignment statements, conditional statements, expressions, string and character handling, data validation examples. Iteration, arrays, strings processing. Defining function, function prototype, passing parameters, recursion.

UNIT - II

Pointers: Definition and uses of pointers, pointer arithmetic, pointers and array, pointers and functions, pointer to pointer. Structures, union, pointers to structures, user-defined data types, enumeration. Introduction to Dynamic Memory Allocation, command line arguments, systems calls. File Handling in C

UNIT - III

OOP Paradigm: Characteristics of OOP, Comparison between functional programming and OOP approach, characteristics of object oriented language - objects, classes, inheritance, reusability, user defined data types, polymorphism, overloading., Introduction to C++, Identifier and keywords, constants, C++ operators, type conversion, Variable declaration, statements, expressions, input and output, conditional expression loop statements, break control statements.. Classes, member functions, objects, arrays of class objects, pointers and classes, nested classes, constructors, destructors, Inline member functions, static class member, friend functions, and dynamic memory allocation.

UNIT - IV

Polymorphism and Inheritance: Function overloading, operator overloading, polymorphism, early binding, polymorphism with pointers, virtual functions, late

binding, pure virtual functions. Single inheritance, types of inheritance, types of base classes, types of derivations, multiple inheritances, container classes, member access control.

UNIT - V

Exceptions and Templates: Exception syntax, Multiple Exceptions, Function templates, function templates with multiple argument templates. File Handling in C++

Text Book:

1. E Balagurusamy, "Object oriented Programming with C++", Tata McGraw Hill, 2006
2. Kamthane, A.N., "Programming with ANSI and Turbo C", Pearson Education, Delhi, 2006

References

1. Deitel and Deitel "C How to Program", Pearson Education. 2010
2. Yashavant Kanetkar "Understanding Pointers in C", BPB Publications, 2008
3. Reams Thareja, "Programming in C" Oxford University Press, 2011
4. HM Deitel and PJ Deitel "C++ How to Program", Prentice Hall, 2010
5. Kamthane," Object Oriented Programming with ANSI and Turbo C++",Pearson Education, 2003.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

First Semester

Branch	Subject Title	Subject Code
MCA	Operating System	MCA103

UNIT - I

Introduction : Definition and types of operating systems, Batch Systems, multi programming, time- sharing, parallel, distributed and real-time systems, Operating system structure, Operating system components and services, System calls, system programs, system boot.

UNIT - II

Process Management : Process concept, Process scheduling, Cooperating process, Threads, Inter- process communication, CPU scheduling criteria, Scheduling algorithms, Multiple-processor scheduling and Algorithm evaluation.

UNIT - III

Process synchronization and Deadlocks : Critical-Section problem, synchronization hardware, Semaphores , Classical problem of synchronization, Critical regions, Monitors, Deadlock-system model, Characterization, Deadlock prevention, Avoidance and Detection, Recovery from deadlock, Combined approach to deadlock handling.

UNIT - IV

Storage Management: Memory Management –Logical and Physical Address Space, Swapping, Contiguous Allocation, Paging, Segmentation with paging, Virtual Memory, Demand paging and its performance, Page replacement algorithms, Allocation of frames, Thrashing , Page Size and other considerations.

UNIT - V

Protection and Security: Goals of protection, Domain of protection, Access matrix, Implementation of access Matrix, Revocation of Access Rights, Language based protection, The security problem, authentication, One Time passwords, Program threats, System threats, Threat Monitoring, Encryption.

UNIT - VI

File system and secondary storage structure: file concept, access methods, directory implementation, Efficiency and performance, recovery, Disk structure, Disk scheduling methods, Disk management, Recovery, Swap-space management, Disk reliability.

UNIT - VII

Case Study: Windows NT-Design principles, system components Environmental Subsystems, File system, Networking and program interface.

Text Book:

1. Abraham Silberschalz, Peter B Galvin and G. Gagne, "Operating System Concepts", Addison Wesley Publishing Co. , 2010

References:

1. Andrew S.Tanenbaum, "Modern Operating System", PHI Learning Pvt. Ltd., 2008
2. William Stallings, "Operating Systems : Internal and design Principles", Prentice Hall, 2011
3. H.M. Deital, PJ Deital and DR Chorffines, "Operating System", Pearson Education, 2011



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

First Semester

Branch	Subject Title	Subject Code
MCA	Computer Architecture	MCA104

UNIT - I

Basic Building Blocks: Gates, Boolean Functions and Expressions Designing Gate Networks, K-map simplification, Useful Combinational Parts, Programmable Combinational Parts, Timing and Control, Latches, Flip-flops, Registers and Counters, Sequential Circuits.

UNIT - II

Arithmetic/Logic Unit: Numbers Representation, Arithmetic Operations, Floating-Point Arithmetic.

UNIT - III

Register Transfer Language and Micro-operations: Concept of bus, data movement among registers, a language to represent conditional data transfer, data movement from/ to memory. Design of Arithmetic & Logic Unit and Control Unit Control design hardwired control, micro programmed arithmetic and logical operations along with register transfer, timing in register.

UNIT - IV

Instruction and Addressing: A simple computer organization and instruction set, instruction formats, addressing modes, instruction cycle, instruction execution in terms of microinstructions, interrupt cycle, concepts of interrupt and simple I/O organization, Synchronous & Asynchronous data transfer, Data Transfer Mode: Program Controlled, Interrupt driven, DMA(Direct Memory Access). Implementation of processor using the building blocks.

UNIT - V

Memory System Design: Memory Origination, Memory Hierarchy, Main Memory (RAM/ROM chips), Auxiliary memory, Associative memory, Cache Memory, Virtual Memory. Assembly Language Programs, Assembler Directives, Pseudo Instructions, Macroinstructions, Linking and Loading.

UNIT - VI

Vector and Array Processing: Shared-Memory, Multiprocessing, Distributed Multi Computing.

UNIT - VII

Microprocessor Concepts: Pin Diagram of 8085, Architecture of 8085, Addressing Mode of 8085, functional block diagram of 8085 assembly language, instruction set of 8085.

Text Book:

1. M. Morris Mano "Computer System Architecture" Prentice Hall , 2007

References:

1. William Stallings, "Computer Organization & Architecture - Designing for Performance", 2012
2. John P~ Hayes, "Computer Architecture and Organization" , Tata McGraw Hill, 1998.
3. Svetlana N.Yanushkevich, Vlad P.Shmerko, "Introduction to Logic Design", CRC Press, 2012.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

First Semester

Branch	Subject Title	Subject Code
MCA	Accounting and Financial Management	MCA105

UNIT - I

Introduction to Accounting: Definition of Accounting and its advantages & limitations, Scope of accounting. Branches of Accounting - Financial Accounting - Cost Accounting - Management Accounting, users of Accounting information, Methods of Accounting, Double Entry Accounting System, Types of Accounts and Rules for Debit and Credit Preparation of Journal, Ledger and Trial Balance

UNIT - II

Final Accounts: Preparation of Final Accounts (Sole Proprietorship only), Preparation of Trading A/c, Profit & Loss A/c and Balance Sheet covering simple adjustments.

UNIT - III

Accounting Ratios: Meaning, Advantages and Limitations of Accounting ratios, Computation of profitability, liquidity, solvency, and turnover ratios.

UNIT - IV

Cost Accounting: Meaning and definition of Cost Accounting – its Advantages & Limitations, Marginal Costing: Meaning-Advantages- Limitations, Break Even Point, Margin of Safety, Profit Volume Ratio

UNIT - V

Introduction to Financial Management: Meaning of financial management, functions and

objectives of financial management- Profit Maximization and wealth maximization, basic finance decisions, changing role of finance.

Text Books:

1. S.N. Maheswari, & S.K. Maheshwari, “Advanced Management Accounting”, Vikas Publishing House, 2010
2. M.Y. Khan and P.K. Jain, “Management Accounting”, Tata McGraw Hill, 2009
3. M Y. Khan and P.K. Jain , “Cost Accounting”, Tata McGraw Hill, 2008

References:

1. Vijay Kumar , “Accounting for Management”, Tata McGraw Hill, 2015
2. Anthony, N. Robert, Hawkins and Merchant , “Accounting: Text and Cases(SIB)”, McGraw Hill, 2013
3. Weston, John. Fred. Thomas, E.Copeland , “Managerial finance”, Dryden press, 1988



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

First Semester

Branch	Subject Title	Subject Code
MCA	Office management Lab	MCA151

MS word Basics: Introduction to MSWord; Features & area of use. Working with MS Word. Menus & Commands, Toolbars & Buttons, Shortcut Menus, Wizards & Templates. Creating a New Document; Different Page Views and layouts; Applying various Text enhancements. Working with - Styles, Text Attributes; Paragraph and Page Formatting; Text Editing using various features; Bullets, Numbering, Auto formatting, Printing & various print options.

Advanced Features of MS-Word: Spell Check, Thesaurus, Find & Replace; Headers & Footers, Inserting - Page Numbers, Pictures, Files, Auto texts, Symbols etc., Working with Columns, Tabs & indents; Creation & Working with Tables including conversion to and from text; Margins & Space management in Document; Adding References and Graphics; Mail Merge, Envelops & Mailing Labels.

MS Excel: Introduction and area of use; Working with MS Excel., concepts of Workbook & Worksheets; Using Wizards; Various Data Types; Using different features with Data, Cell and Texts; Inserting, Removing & Resizing of Columns & Rows; Working with Data & Ranges; Different Views of Worksheets; Column Freezing, Labels, Hiding, Splitting etc.; Using different features with Data and Text; Use of Formulas, Calculations & Functions; Cell Formatting including Borders & Shading; Working with Different Chart Types; Printing of Workbook & Worksheets with various options

MS PowerPoint: Introduction & area of use; Working with MS PowerPoint; Creating a New

Presentation; Working with Presentation; Using Wizards; Slides & its different views; Inserting, Deleting and Copying of Slides; Working with Notes, Handouts, Columns & Lists; Adding Graphics, Sounds and Movies to a Slide; Working with PowerPoint Objects; Designing & Presentation of a Slide Show; Printing Presentations

MS Access: Define data needs and types ,define and print table relationships, Add, set, change or remove primary keys Split databases Create databases Create tables Modify tables Enter records using datasheet view, Delete records from a table Change records in a table Create fields and modify field properties Create reports Modify the design of reports and forms.

Computerized Accounting :Use of Accounting Software Tally, Creation Company, Voucher Entry, Types of accounts, Mode of accounting (Day cash book, ledger book, bank reconciliation, Ledgers, Trial balance, Balance Sheet) Analysis of Trial Balance and Final Accounts



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

First Semester

Branch	Subject Title	Subject Code
MCA	C & C++ Lab	MCA152

Control Statements:-Programs on if-else ladder, iterative statements, Functions and recursions, predefined functions.

Pointer and Dynamic Memory:-Programs on Arrays, sorting (Bubble, selection, insertion) Searching (linear, Binary), 2D Array (Matrix operations), Pointers, Structures, union, enum, Dynamic Memory allocation. Programs on File Handling, Programs on Command Line Arguments.

Objects, Functions and Constructor:- Programs on classes and objects constructors, functions , inline functions, Friend function.

Polymorphism:-Programs on Function Overloading, overriding, Operator overloading, programs on different type of inheritances, virtual function.

Exception Handling and File Handling:- Programs on input/output Streams, Exception Handling, File Handling ,Template Classes.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

First Semester

Branch	Subject Title	Subject Code
MCA	Communication Skills	MCA153

Listening and Speaking Skills: Conversational Skills Formal and Informal Conversation listening to lectures, discussions, talk shows, news programs.

Writing Skills: Formal & Informal writings, report writing, Creative writing seminar, paper, bibliography.

Composition: Resume Writing, Business Letter Writing: Complaint, Job Applications and official correspondence; E-mail messages.

Presentation Skills: Elements of effective presentation, structure of presentation, voice modulation, audience analysis, body language, eye contact, sample videos. Debates, Seminar, Speeches, Lectures. Interviews, mock interviews.

Group Discussion: Structure of GD, Moderator led and other ODs, Strategies in GD, team work body language, mock GD



RKDF UNIVERSITY, RANCHI

New Scheme of Examination as per AICTE Flexible Curricula

MCA(Master in Computer Application)

SCHEME

Second Semester

S.No.	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
			Max	Max	Min	Max	Min
1	MCA201	Data Structures	30	70	21	100	35
2	MCA202	Java Programming	30	70	21	100	35
3	MCA203	Database Management System	30	70	21	100	35
4	MCA204	Enterprise Resource Planning	30	70	21	100	35
5	MCA205	Web Programming	30	70	21	100	35
Practical							
Total							
			Max		Min		
1	MCA251	Data Structures Lab	50		25		
2	MCA252	Java Programming Lab	50		25		
3	MCA253	RDBMS Lab	50		25		
4	MCA255	Web Programming Lab					



RKDF UNIVERSITY, RANCHI

New Scheme of Examination as per AICTE Flexible Curricula

MCA(Master in Computer Application)**SCHEME****Second Semester**

MCA Year 1 Semester 2						
S.No.	Subject Code	SUBJECT NAME	PERIODS			Credit
			L	T	P	
1	MCA201	Data Structures	3	1	0	4
2	MCA202	Java Programming	3	1	0	4
3	MCA203	Database Management System	3	1	0	4
4	MCA204	Enterprise Resource Planning	3	1	0	4
5	MCA205	Web Programming	3	1	0	4
Practicals						
1	MCA251	Data Structures Lab			2	1
2	MCA252	Java Programming Lab			2	1
3	MCA253	RDBMS Lab			2	1
4	MCA255	Web Programming Lab			2	1
Total					8	24



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Second Semester

Branch	Subject Title	Subject Code
MCA	Data Structures	MCA201

UNIT - I

Introduction: Basic data structures such as arrays, linked list, stack, trees and queues and their applications, linked and sequential representation Basic Terminology, Elementary Data organization, Data Structure operations.

UNIT - II

Queues, Stack implementation through Arrays: Array Definition, Representation and Analysis, Single and Multidimensional Arrays, address calculation (row major and column major), Applications of arrays - Character Strings and string operations, Ordered Lists.

Stack: Implementation of stack, operations on stack. Applications of stack: Conversion of infix-expressions to prefix and postfix expressions, evaluation of postfix expression.

UNIT - III

Queues: implementation of queues, Operations on Queue, Types of Queues - Circular queue, Deque and Priority Queue.

Recursion: recursion in C, example of recursion, Tower of Hanoi Problem, simulating recursion, Backtracking,, recursive algorithms, principles of recursion.

Linked list: Representation and Implementation of Singly Linked Lists, Two-way Header List, Traversing and Searching of Linked List, insertion and deletion to/from Linked Lists, insertion and deletion Algorithms, Doubly linked list, Header lists, circular lists, sorted lists.

UNIT - IV

Trees: Basic terminology and definitions. Array and Linked Representation of Binary trees, Traversing Binary trees.

Binary Search Trees: Binary Search Tree (BST), Traversal, Insertion and Deletion in BST, and Introduction to balanced BST (AVL Trees).

Searching: Sequential search, binary search, comparison and analysis.

Sorting: Insertion Sort, Bubble Sort, Quick Sort, Two-Way-Merge Sort, Heap Sort, Sorting on Different Keys, Practical consideration for internal Sorting.

UNIT - V

Graphs: Terminology & Representations, Graphs & Multi-graphs, Directed Graphs, Sequential Representations of Graphs, Adjacency Matrices, Traversal, Connected Component and Spanning Trees, Minimum Cost Spanning Trees

Text Book:

1. Abraham Silberschalz, Peter B Galvin and G. Gagne, “Operating System Concepts”, Addison Wesley Publishing Co. , 2010

References:

1. Andrew S.Tanenbaum, “Modern Operating System”, PHI Learning Pvt. Ltd., 2008
2. William Stallings, “Operating Systems : Internal and design Principles”, Prentice Hall, 2011
3. H.M. Deital, PJ Deital and DR Chorffnes, “Operating System”, Pearson Education, 2011



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Second Semester

Branch	Subject Title	Subject Code
MCA	Java Programming	MCA202

UNIT - I

Fundamentals of Java Programming: Data in Java Programs, Arithmetic Operators and Expressions, Simple Program Input and Output. Making Decisions with Java: Comparing Numbers in Java, Comparing Strings in Java, Logical (Boolean) Operators and Order of Precedence, Selection Structures in Java.

UNIT - II

Repeating Program Statements: The while Statement, The for Statement, The do...while Statement, Nested Loops, break, and continue. Methods and Classes: Predefined Java Methods, Programmer-Defined Methods. Arrays: The One-Dimensional Array, Multidimensional Arrays, Other Array Topics.

UNIT - III

Characters, Strings, and Formatting: Working with Characters, Working with Strings, Formatting Data for Output. Exceptions and Assertions: Exception Handling, Assertions

UNIT - IV

File Input and Output: Inputting Data from a Text File, Outputting Data to a Text file, Performing Input and Output with Binary Files.

UNIT - V

Graphical User Interfaces: Creating User Interfaces, Overview of a Java GUI, Developing a Java GUI, Adding Functionality to a GUI, Improving GUI Layout.

Text Book:

1. Johnson Richard A., "An Introduction to Java Programming and Object-Oriented Application Development", 1st Edition, Thomson Learning, New Delhi, 2007.

Reference Books:

1. Dietel&Dietel, "Java How to Program", 5th Edition, Pearson Education, New Delhi, 2006.
2. Balagurusamy E. "JAVA Programming", 3rd Edition, TMH, New Delhi, 2005.
3. Sleek James M. Sleek "Programming and Problem Solving with JAVA", Thomson Learning, Indian Edition, 2007.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Second Semester

Branch	Subject Title	Subject Code
MCA	Database Management System	MCA203

UNIT - I

Introduction: Overview of DBMS, Advantages of DBMS, Basic DBMS terminology, Database System v/s File System, Data Independence, Architecture of DBMS, Introduction to data models: Relational Model, Network Model, Hierarchical Model, Entity-Relationship Model, Comparison of network, hierarchical and relational models.

UNIT - II

Data modeling using the Entity Relationship Model: ER model concepts, Types of Relationships, notation for ER diagram, Reduction of ER-Diagrams to Relational Model, mapping constraints, Generalization, Aggregation, Specialization, Extended ER model, relationships of higher degree

UNIT - III

Relational model: Storage Organizations for Relations, Relational Algebra, Set Operations, Relational Calculus, Concepts of Alternate key, candidate key, primary key, Foreign key, Integrity Rules, Data Dictionary.

UNIT - IV

Normalization: Functional dependencies, normal forms, first, second, third normal forms, BCNF, inclusion dependencies, loss less join decompositions, normalization using FD, MVD, and JDs, alternative approaches to database design.

UNIT - V

Introduction to SQL: Characteristics of SQL, Advantages of SQL, SQL data types and literals, Types of SQL commands: DDL, DML, TCL, DCL, SQL operators, Tables, views and indexes, Constraints Group By and Having Clause, Order By Clause, Queries and sub queries, Aggregate Functions, Numeric Functions, String Functions, Date & Time Functions, Insert, Update and Delete operations, Unions, Intersection, Minus, Joins: Equi-Join, Natural Join, Self-Join, Inner Join, Outer Join

Text Books:

1. Elmasri, Navathe, “Fundamentals of Database Systems”, Addison Wesley, 6th Edition, 2011
2. Korth, Silberschatz, Sudarshan, “Database Concepts”, McGraw Hill, 6th Edition, 2010

References

1. Thomas Connolly and Carolyn Begg, “Database Systems: A Practical Approach to Design, Implementation, and Management, Addison Wesley, 5th Edition, 2010
2. Ramakrishnan, Gehrke, “Database Management System”, McGraw Hill, 3rd Edition, Jan 2007
3. Date C J, “An Introduction to Database System”, Addison Wesley, 8th Edition 2003
4. Bipin C. Desai, “An Introduction to Database Systems”, Galgotia Publication, Revised Edition, 2010
5. Majumdar & Bhattacharya, “Database Management System”, TMH, 2005
6. Paul Beynon Davies, “Database Systems”, Palgrave Macmillan, 3rd Edition, 2003



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Second Semester

Branch	Subject Title	Subject Code
MCA	Enterprise Resource Planning	MCA204

UNIT - I

Introduction To ERP

Overview, Benefits of ERP, ERP and Related Technologies, Business Process Reengineering, Data Warehousing, Data Mining, On-line Analytical Processing, Supply Chain Management

UNIT - II

ERP Implementation

Implementation of Life Cycle, Implementation Methodology, Hidden Costs, Organizing Implementation, Vendors, Consultants and Users, Contracts, Project Management and Monitoring

UNIT - III

Business Modules

Business Modules in an ERP Package, Finance, Manufacturing, Human Resource, Plant Maintenance, Materials Management, Quality Management, Sales and Distribution

UNIT - IV

ERP Market

ERP Market Place, SAP AG, PeopleSoft ,Baan Company , JD Edwards World Solutions Company, Oracle Corporation, QAD , System Software Associates.

UNIT - V

ERP–Present AndFuture

Turbo Charge the ERP System, EIA, ERP and E–Commerce, ERP and Internet, Future Directions in ERP.

Books:-

1. Alexis Leon, “ERP Demystified”, TMH.
2. V. K. Garg & N. K. Venkita Krishnan, “ERP Ware: ERP Implementation Framework”.
3. V. K. Garg & N. K. Venkita Krishnan, “ERP Concepts & Planning”.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Second Semester

Branch	Subject Title	Subject Code
MCA	Web Programming	MCA205

UNIT - I

MVC, Asp.NET MVC, ORMs, Entity Framework, Models, Database Contexts, Adding Controllers, Views, Filtering, Searching related entities.

UNIT - II

ViewBag, View Model, Complex Filtering, Data Validation, Annotations, Sorting, Paging, Routing Configurations, Many to many relationships with the Entity Framework, Partial Views.

UNIT - III

Authentication, Authorization, ASP.NET Identity, Role Management, User management, Password management.

UNIT - IV

CSS Fundamentals, Selectors, Inheritance, Cascading, Box Model, Advanced CSS, Animations.

UNIT - V

Designing Responsive web sites, Media Queries, Developing for mobiles and Tablets.

Text books:

1. Naylor L., "ASP.NET MVC with Entity Framework and CSS", 1st Edition, Apress, 2017.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Second Semester

Branch	Subject Title	Subject Code
MCA	Data Structures Lab	MCA251

1. Program to Find the Number of Elements in an Array
2. Develop and Implement a menu driven program in C for the following Array operations a. Creating Array of N Integer elements. b. Display of Array elements with suitable headings. c. Inserting an element (ELEM) at a given valid position (POS). d. Deleting an element at a given valid position (POS). e. Exit
3. Programs for Stack, Queues and Circular Queues using Arrays
4. Program to convert an Infix Expression into Postfix and Postfix Evaluation
5. Program to implement stack using arrays
6. Program to implement stack using linked list
7. Program to implement multiple stack in a single array
8. Program to convert infix notation to postfix notation using stacks
9. Program to implement queue using arrays
10. Program to implement queue using pointers
11. Program to reverse elements in a queue
12. Program to implement circular queue using arrays
13. Program to create add remove & display element from single linked list
14. Program to create add remove & display element from double linked list
15. Program to count number of nodes in linear linked list
16. Program to create add remove & display element from circular linked list
17. Programs to implement stack & queues using linked representation
18. Program to concatenate two linear linked lists
19. Program to accept a singly linked list of integers & sort the list in ascending order.
20. Program to reverse linked list
21. Program to represent polynomial using linked list
22. Program to add two polynomials using linked list
23. Program for the creation of binary tree, provide insertion & deletion in c
24. Program for pre-order, post-order & in-order traversals of a binary tree using non recursive.
25. Program to count no, of leaves of binary tree
26. Program for implementation of B-tree (insertion & deletion)
27. Program for implementation of multi-way tree in c
28. Program for implementation of AVL tree
29. Program to implement bubble sort program using arrays
30. Program to implement merge sort using arrays

31. Program to implement selection sort program using arrays
32. Program to implement insertion sort program using arrays
33. Program to implement topological sort using arrays
34. Program to implement heap sort using arrays
35. Program to implement heap sort using pointers
36. Program to implement bubble sort program using pointers
37. Program to implement linear search using pointers
38. Program to implement binary search using pointers
39. Program to implement linear search using arrays
40. Program to implement binary search using arrays

Text books:

1. Baluja G S, “Data Structure through C”, Ganpat Rai Publication, New Delhi, 2015.
2. Pai G A V, “Data Structures and Algorithms: Concepts, Techniques and Applications”, 2 ndEdn, Tata McGraw-Hill, 2008.
3. Horowitz E., Sahni S., Susan A., “Fundamentals of Data Structures in C”, 2nd Edition, University Press, 2010.

Reference books:

1. Tremblay J. P., Sorenson P. G, “An Introduction to Data Structures with Applications”, 2nd Edn, McGraw-Hill, Inc. New York, NY, USA.
2. Lipschutz Seymour, “Data Structures”, 6th Edn, 9th Reprint 2008, Tata McGraw-Hill.
3. Drozdek Adam, “Data Structures and Algorithms in C++”, Thomson Learning, New Delhi – 2007.
4. Feller J., Fitzgerald B., “Understanding Open Source Software Development”, Pearson Education Ltd. New Delhi



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Second Semester

Branch	Subject Title	Subject Code
MCA	Java Programming Lab	MCA252

1. WJJP to show the characteristic of a number. {E.g. 24 it has two coefficients 2 in tens position and 4 in units position. It is composed of 2 and 3. It is a positive number. Also show whether it is odd or even.
2. WJJP to take input through command line argument and do the following: a) Check whether the number is prime. b) Generate the reverse a number.
3. Write a menu driven program using switch in Java to perform following: a) For input of 1, check whether the number is prime b) For input of 3, find the factors of the number c) For input of 5, check the number is odd or even.
4. Write a program in Java to generate hexadecimal equivalent of a number without using array.
5. WJJP to take two number inputs through command line argument and do the following: a) Check whether two numbers are prime to each other or not. b) Find LCM of two numbers.
6. WJJP to create a class and exhibit the role of static functions (other than main) by declaring, defining and calling them.
7. WJJP to compute and display the count of occurrence of 4 in a number. E.g. 4564 will compute 2.
8. WJJP to take an angle value in degrees and then compute the equivalent radians and then prove $\sin^2 \theta + \cos^2 \theta = 1$. Note $180^\circ = \pi^c$
9. WJJP to sort a list of numbers in ascending order.
10. WJJP to generate Pascal's Triangle using a square matrix.
11. Write a program in Java to take input of two 3×3 matrices through command line argument and then: a) Add them up and display the result b) Subtract them and display the result c) Multiply them and display product
12. WJJP to count the number of words, characters in a sentence.
13. Write a program in Java to take input of a sentence through command line argument and then count the number of words and vowels.
14. WJJP to handle the Exception using try and multiple catch block; the exceptions that you will handle are, number format error, array bound error and divide by zero.
15. WJJP to create a class called Room with two data member length and width and then implement constructor overloading in it.
16. Write a program in Java to explain the role of the following: a) Non-parameterized constructor b) Parameterized constructor c) Copy constructor Take input and display the output.

17. WAJP to create a class called Fraction with data member numerator and denominator; take input (through command line argument) of two fractions and then add, subtract, multiply and divide, finally display the result in reduced term.
18. Write a program in Java to create a class for Employee having 2 data member code and name. Then create 3 classes Officer, AdminStaff and MStaff. The Officer class has data members designation and pay-scale; the AdminStaff has data members grade and pay-band; the MStaff has data member department and two sub-classes Regular and Casual. The Regular staff has data members level and consolidated-pay and Casual has data member daily-wage. Take all inputs through constructors and write appropriate methods for displaying one data for each type of class.
19. WAJP to design a class called Account using the inheritance and static that show all function of bank (withdrawal, deposit) and generate account number dynamically.
20. WAJP to design an application Password.java that produces and prints a random password depending upon name of an individual. If the input is Abdul Kalam then the password would be 33421LAM. Note: take the first name A=1, B=2, D=4, U=21 where 2+1=3, and L=12, where 1+2=3; so the number comes to be 12433, so u can find out.
21. WAJP to draw a format like

```

      *
    ***
  *****
*****
  *****
    ***
      *

```

22. WAJP to take a string count all vowels and then delete the same from the string.
23. Write a Patient class which inherits from the Person class. Patient can again be of two types, indoor and outdoor. The Patient class requires the following: a) a variable to store the patient ID for the patient b) a variable to store the department of hospital c) a variable to store the ward of hospital d) a variable to store the patient 's date of joining the hospital e) a variable to store the patient 's address f) a variable to store the medical fees that the patient pays g) constructor methods, which initialize the variables h) a method to calculate the medical fees (for both indoor and outdoor patient)
24. WAJP to take a string as password and check whether it contains at least two numbers, 3 alphabets and no space in it. If any contrary throw message.
25. Write a program in Java to create a class called Rational having two data members for numerator and denominator. Take two inputs of rational numbers and perform multiplication and division. Display the result in reduced form.
26. Write a program in Java to print a format like,

```

* * * * *
* * * *
* * *
*

```

27. Write a class called Shape which contains a user-defined interface for

Computation, which contains methods for calculation of area, perimeter and volume. Write four classes for circle, rectangle, sphere and rectangular parallelepiped, and all these classes inherit from Shape. Now take input for the following:

- a) radius of circle and compute its area and perimeter
- b) Length and breadth of rectangle and compute its area and perimeter
- c) Length, breadth and height for rectangular parallelepiped and compute its area and volume
- d) Radius of sphere and compute its area and volume

** Area of circle= πr^2 , perimeter of circle= $2\pi r$, area of sphere= $4\pi r^2$, volume of sphere= $\frac{4}{3}\pi r^3$, volume of rectangular parallelepiped = $h \times b \times l$ area of rectangular parallelepiped= $2(l \times h + h \times b + b \times l)$

28. Write a class called Employee, which requires the following: a) a variable to store the employee ID a. employee ID should be of format EMPM1234, EMPS1234, EMPA1234, EMPC1234, where M=manager, S=supervisor, A=analyst, C=clerk; number can be any no. but first three characters should be EMP b) a variable to store the employee name c) a variable to store department d) a variable to store city e) a variable to store basic salary f) a method to calculate the salary of employee a. if the city is metro then the HRA would be 30% else 20% b. if the employee ID contain M then DA would be 120%, if S then DA would be 110%, if A then DA would be 100%, and if C then DA would be 90% g) constructor methods, which initialize the variables
29. WAP to create 4 threads and show exhibit their execution after the call of the "start ()" method.
30. Write a program in Java to create 3 threads and exhibit their behaviour by changing their priorities in the "main" thread. Display the possible output.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Second Semester

Branch	Subject Title	Subject Code
MCA	RDBMS Lab	MCA253

Consider the following tables:

emp(empno,ename,job,mgr,hiredate,sal,comm,deptno,gr),

dept(deptno,dname,loc)

Write the following queries:

1. List all information about all department from emp table.
2. List all employee names along with their salaries from emp table.
3. List all department numbers, employee numbers and their managers numbers in descending order of deptno from emp table.
4. List department names and locations from the dept table.
5. List the employees belonging to the department 20.
6. List the name and salary of the employees whose salary is more than 1000.
7. List the names of the clerks working in the department 20.
8. List the names of analysts and salesmen.
9. List the details of the employees who have joined before the end of September 81.
10. List the names of employees who are not managers.
11. List the names of employees whose employee number are 7369, 7521, 7839, 7934, 7788.
12. List the employee details not belonging to the department 10, 30, and 40.
13. List the employee name and salary, whose salary is between 1000 and 2000.

14. List the employee names, who are not eligible for commission.(salary having >15,000 eligible for commission)
15. List the employees who are eligible for commission.
16. List the details of employees, whose salary is greater than 2000 and commission is NULL.
17. List the employees whose names start with an “S” (not”s”).
18. List the name, salary and PF amount of all the employees(PF is calculated as 10% of salary).
19. List the empno, ename, sal in ascending order of salary.
20. List the employee name, salary, job and Department no descending order of Department No and salary.
21. List the employee details in ascending order of salary.
22. List the employee details in descending order of salary
23. Display name, and sal and commission of all employees whose monthly salary is greater than their commission.
24. Select SMITH HAS WORKED IN THE POSITION OF CLERK IN DEPT 20.Display result in this format.
25. Generate a statement which prompts the user at runtime. The intention is to display employees hired between 2 given dates.
26. Define a variable representing an expression used to calculate total annual remuneration. Use the variable in a statement which finds all employees who earn \$30000 a year or more.
27. List all the employees name and salaries increased by 15% and expressed as a whole number of dollars
28. Produce the following EMPLOYEE AND JOB SMITH CLERK ALLEN SALESMAN
29. Produce the following output: SMITH (Clerk) ALLEN (Salesman)
30. Do a case sensitive search for a list of employees with a job that the user enters.
31. It has been discovered that the sales people in dept. 30 are not all male. Please produce thefollowingoutput. ENAME DEPTNO JOB ALLEN 30 Sales Person

32. Display each employees name and hiredate of dept 20.
33. Display each employees name, hiredate and salary review date. Assume salary review date is one year from hiredate. Output should be in ascending review date.
34. Print list of employees displaying just salary, if more than 1500. If exactly 1500 display “ On Target”. If less than 1500 display “ Below 1500”.
35. Write a query which returns DAY of the week (i.e. MONDAY) for any date entered in the format DD/MM/YY.
36. Write a query to calculate length of service of each employee.
37. Find the minimum salary of all employees.
38. Find the maximum, minimum, and average salaries of all employees.
39. List the maximum and minimum salary of each job type.
40. Find how many managers are in each dept.
41. Find the average salary and average total remuneration of each job type. Remembers sales man earn commission.
42. Find out the difference between highest and lowest salary.
43. Find all department s which have more than three employees.
44. Check whether all employee nos are unique. (No Duplicate)
45. List lowest paid employee working for each Manager. Exclude any groups where the minimum salary is less than 1000. Sort the output by salary.
46. Produce a list showing employees ‘salary grade’.(> 10000 A, >10000 &20000 C)
47. Show only employee on Grade C.
48. .Show all employee in Dallas.
49. List the employees name, job, salary, grade and department for everyone in the company except clerks. Sort on salary, displaying the highest first.
50. List the following details of employees who earn \$36000 a year or who are clerks.
Ename Job Annual Sal Dept no Dname Grade
51. Display all employees who earn less than their managers.
52. Display all employees by name and eno along with their managers name and number.

53. Modify above spoliation to display KING who has no MANAGER.
54. Find the job that was files in the first half of 1983 and the name job that was filled in the same period in 1984.
55. Find all employees who have joined before their manager. EMPLOYEE HIREDATE
MANAGER HIREDATE
56. Find the employees who earn the highest salary in each job, type, sort in descending order of salary.
57. Find the employees who earn the minimum salary for their job, Display the result in descending order of salary
58. Find the most recently hired employees in the department. Order by hiredate.
59. Show the details of any employee who earns a salary greater than the average for their department. Sort in department number order.
60. List all department where there are no employees.

Text book:

1. SQL, PL/SQL the programming Language of Oracle, Ivan Bayross, 4th edition



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Second Semester

Branch	Subject Title	Subject Code
MCA	Web Programming Lab	MCA255

List of Programs as Assignments:

1. To create a simple html file to demonstrate the use of different tags.
2. To create an html file to link to different html page which contains images, tables, and also link within a page.
3. To create an html page with different types of frames such as floating frame, navigation frame & mixed frame.
4. To create a registration form as mentioned below. Create an html page named as “registration.html” a) set background colors b) use table for alignment c) provide font colors & size 2.
5. To create an html file by applying the different styles using inline, external & internal style sheets.
 - Create a external style sheet named as “external_css.css” and provide some styles for h2, hr, p & a tags.
 - Create an html file named as “Style_sheet.html” a) Include the external style sheet with necessary tag. b) Include the internal style sheet for body tags & also use class name, so that the style can be applied for all tags. c) Include a tags with inline style sheet.
6. To write a Javascript program to define a user defined function for sorting the values in an array.
7. Create an html page named as “exception.html” and do the following. 1. within the script tag write code to handle exception a) define a method RunTest() to get any string values(str) from the user and cll the method Areletters(str). b) In Areletters(str) method check whether str contain only alphabets (a-z, AZ), if not throw exception. Master of Computer Applications167 c) Define a exception method Input Exception(str) to handle the exception thrown bythe above method. 2. Within the body tag define a script tag to call Runtest() method defined.
8. To display the calendar using javascript code by getting the year from the user.
9. To create a html page to display a new image & text when the mouse comes over the existing content in the page.
10. To create an ASP file to find the no of hits on the page and to have rotating banner content.
11. To create a table of content using ASP program & navigate within the pages.
12. Create an ASP file named as request.asp a) Create a simple form to get the first name & last name and a button submit. When the button is clicked the values in

- the text box are printed by response object by Request.QueryString b) Create a hyperlink with some values defined in the tag & display the same using request & response object.
13. To display all the content in the database using ASP program.
 14. To create a simple servlet program to display the date (using Tomcat server).
 15. To create a servlet program to retrieve the values entered in the html file (Using NetBeans IDE).
 16. To display the cookie values that are entered in the html page using servlet program. (using NetBean IDE).
 17. To create a simple catalog using XML file
 18. To create external style sheet and using the style sheet in xml file.
 19. To create a php program to demonstrate the different file handling methods.
 20. To create a php program to demonstrate the different predefined function in array, Math, Data & Regular Expression.

TEXT BOOKS

1. Web Technologies: A Computer Science Perspective , Jeffrey C Jackson , Pearson Education , India.
2. Stephen Wynkoop, Running a perfect website, QUE, 1999

REFERENCE BOOKS

1. Eric Ladd, Jim O' Donnel, Using HTML 4, XML and Java, Prentice Hall of IndiaQUE, 1999
2. Chris Bates, Web Programming - Building Intranet applications, Wiley Publications, 2004
3. Deitel, Deitel& Nieto, Internet and World Wide Web - How to Program, Pearson Education Asia, 2000



RKDF UNIVERSITY, RANCHI

New Scheme of Examination as per AICTE Flexible Curricula

MCA(Master in Computer Application)

SCHEME

Third Semester

S.No.	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
			Max	Max	Min	Max	Min
1	MCA301	Software Engineering	30	70	21	100	35
2	MCA302	Fundamentals of Computer Algorithms	30	70	21	100	35
3	MCA303	Automata Theory	30	70	21	100	35
4	MCA304	Data Mining and Ware Housing	30	70	21	100	35
5	MCA305	Data Communications and Computer Networks	30	70	21	100	35
Practical							
Total							
			Max		Min		
1	MCA351	Software Engineering Lab	50		25		
2	MCA354	Data Mining Lab	50		25		
3	MCA355	Computer Networks Lab	50		25		



RKDF UNIVERSITY, RANCHI

New Scheme of Examination as per AICTE Flexible Curricula

MCA(Master in Computer Application)

Third Semester

Choice Based Credit System

MCA Year 1 Semester I						
S.No.	Subject Code	SUBJECT NAME	PERIODS			Credit
			L	T	P	
1	MCA301	Software Engineering	3	1	0	4
2	MCA302	Fundamentals of Computer Algorithms	3	1	0	4
3	MCA303	Automata Theory	3	1	0	4
4	MCA304	Data Mining and Ware Housing	3	1	0	4
5	MCA305	Data Communications and Computer Networks	3	1	0	4
Practicals						
1	MCA351	Software Engineering Lab			2	1
2	MCA354	Data Mining Lab			2	1
3	MCA355	Computer Networks Lab			2	1
Total					8	23



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Third Semester

Branch	Subject Title	Subject Code
MCA	Software Engineering	MCA301

UNIT - I

Introduction to Software Engineering: Evolving Role of Software, Changing Nature of Software, Legacy Software, Process Frame work, Process Patterns, Process Models, Waterfall Model, Incremental Process Models, Evolutionary Process Models, Specialized Process Models, Unified Process Model, Agile Process Model.

UNIT - II

Requirement Engineering: A bridge to design and construction, Requirement Engineering Task, Initiating the Requirement Engineering Process, Eliciting Requirements, Developing Use case, Building the Analysis Model, Negotiating Requirements, Validating Requirements.

UNIT - III

Design Engineering: Design Process and Design Quality, Design Concepts, Design Models, Pattern Based Software Design

UNIT - IV

Testing Strategies and Testing Tactics: Strategic Approach to software Testing, Test Strategies for conventional and Object Oriented Software, Validation Testing System Testing, White Box Testing, Basic Path Testing Control Structure Testing, Black Box Testing, Object Oriented Testing Methods.

UNIT - V

Metric for process and Estimation Techniques: Process metrics, Software Measurement, Software Project Estimation, Decomposition Techniques, Empirical Estimation Models, Estimation for Object Oriented Projects Specialized Estimation Techniques. Software Quality and Configuration Management: Quality Concepts, Software Quality Assurance,

Software Reliability, Software Configuration Management, SCM Repository, SCM Process.

Text Book:

1. Pressman Roger S., “Software Engineering – A Practitioner’s Approach”, 6th Edition., Tata McGraw Hill.

Reference Books:

1. Vliet Haus Van, “Software Engineering – Principles and Practice”, Wiley John and Sons, 2 nd Edition.
2. Sommerville Ian, “Software Engineering”, 7th Edition., Pearson Education.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Third Semester

Branch	Subject Title	Subject Code
MCA	Fundamentals of Computer Algorithms	MCA302

UNIT - I

Elementary Algorithmic: Introduction, Problems and instances, The efficiency of algorithms, Average and worst-case analyses, What is an elementary operation, why look for efficiency. Asymptotic Notation: Introduction, A notation for “the order of”, Other asymptotic notation, Conditional asymptotic notation, Conditional asymptotic notation, Asymptotic notation with several parameters, Operations on asymptotic notation.

UNIT - II

Analysis of Algorithm: Introduction, Analyzing control structures, Using a barometer, Supplementary examples, Average-case analysis, Amortized analysis, Solving recurrences. Greedy Algorithms: General characteristics of greedy algorithms, Graphs: Minimum spanning trees, Shortest paths, The knapsack problem, scheduling.

UNIT - III

Divide-and-conquer: Introduction: Multiplying large integers, The general template, Binary search, Sorting, Finding the median, Matrix Multiplication, Exponentiation.

UNIT - IV

Dynamic Programming: Calculation the binomial coefficient, The World Series, Making change, The principle of optimality, The knapsack problem, Shortest paths, Chained matrix multiplication.

UNIT - V

Exploring Graphs: Graphs and games: An introduction, Traversing trees, Depth-first search: Undirected graphs, Depth-first search: directed graphs, Breadth-first search, Backtracking, Branch-and-bound, the minimax principle. Probabilistic Algorithms: Introduction, Probabilistic does not imply uncertain, Expected versus average time,

Pseudorandom generation, Numerical probabilistic algorithms, Monte Carlo algorithms, Las Vegas algorithms.

Text Book:

1. Brassard G. & Bratley P., “Fundamentals of Algorithms”, New Delhi, 2005.

Reference Books:

2. 1. E.Horowitz. et.al., “Fundamentals of Computer Algorithms”, Galgotia Publication Pvt. Ltd.,New Delhi, 2004.
3. 2. Kleinberg J. &Tardos E., “Algorithm Design”, Pearson Education, New Delhi, 2006.
4. 3. Cormen T.H., Leiserson Charles E.,Rivest Ronald, Stein Clifford “Introduction to Algorithms” 3rd Edition, PHI, New Delhi, 2005.
5. 4. Dasgupta S., Papadimitriou C.H., Vaziran U.V, “ Algorithm” 3rd Edition , TMH, New Delhi, 2007.
6. 5. Sahani S., “Data Structures Algorithms and Applications in C++” 2 nd Edition, Universities Press (India) Pvt. Ltd., 2005.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Third Semester

Branch	Subject Title	Subject Code
MCA	Automata Theory	MCA303

UNIT - I

Basic Mathematical Objects and Mathematical Induction: Sets, logic, Functions, Relations, Alphabets, Strings, Languages, Principle of mathematical induction, Recursive definition.

UNIT - II

Regular Expressions and Finite Automata: Regular languages and Regular Expressions, Memory required to recognize a language, Finite Automata, capability & limitations of FSM, Deterministic Finite Automata , Non-Deterministic Finite Automata, NFA with e-moves, regular sets & regular expressions, Equivalence of DFA and NFA, NFA from regular expressions, regular expressions from DFA, Moore versus Mealy m/c, two way finite automata equivalence with one way , Kleen's Theorem, applications of finite automata.

UNIT - III

Regular and Non-regular languages: Criterion for Regularity, Minimal Finite Automata, Pumping Lemma for Regular Languages, Decision problems, Regular Languages and Computers. Context Free Grammars: Introduction, definition, Regular Grammar, Derivation trees, Ambiguity, Simplified forms and Normal Forms, Applications.

UNIT - IV

Pushdown Automata: Definition, Moves, Instantaneous Descriptions, Language recognised by PDA, Deterministic PDA, Acceptance by final state & empty stack, Equivalence of PDA , Pumping lemma for CFL, Interaction and Complements of CFL, Decision algorithms. Turing Machines: Definition and examples, Computing Partial Functions with Turing Machine(TM), Combining TMs, Variations of TMs, Multi-tape TMs, Non-deterministic TM, Universal TM, Church Thesis.

UNIT - V

Recursively Enumerable Languages: Recursively Enumerable and Recursive, Enumerating Language, Context Sensitive and Chomsky Hierarchy. Unsolvable Problems and Computable Functions: Nonrecursive Language and unsolvable Problems, Halting Problem, Rice Theorem, Post Correspondence Problem. Computational Complexity: Discussion on P, NP, NPC and NP-Hard Problems.

Text Books:

1. Martin John "Introduction to Languages and the Theory of Computation", 3rd Edition, TMH.

Reference Books:

1. Mishra K.L.P & Chandrasekharan N., "Theory of Computer Science", PHI.
2. Hopcroft John E. And Ullman Jeffrey D., "Introduction to Automata Theory, Languages & Computation", 3rd Edition, Narosa, 2008.
3. Lewis H. R. and Papadimitrou C. H, "Elements of the theory of Computation", PHI.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Third Semester

Branch	Subject Title	Subject Code
MCA	Data Mining and Ware Housing	MCA304

UNIT - I

Data Mining: Introduction, Relational Databases, Data Warehouses, Transactional databases, Advanced database Systems and Application, Data Mining Functionalities, Classification of Data Mining Systems, Major Issues in Data Mining. Data Processing: Data Cleaning, Data Integration and Transformation, Data Reduction.

UNIT - II

Data Warehouse: Introduction, A Multidimensional data Model, Data Warehouse Architecture, Data Warehouse Implementation, Data Cube Technology, From Data Warehousing to Data Mining. Data Cube Computation and Data Generalization

UNIT - III

Mining Association Rules in Large Databases: Association Rule Mining, Single – Dimensional Boolean Association Rules, Multilevel Association Rules from Transaction Databases, Multi Dimensional Association Rules from Relational Databases, From Association Mining to Correlation Analysis.

UNIT - IV

Classification and Prediction: Classification & Prediction, Issues Regarding Classification & Prediction, Classification by decision Tree Induction, Bayesian Classification, Classification by Back propagation, Classification based on concepts & Association Rule Analysis, Other Classification Methods, Prediction, Classification Accuracy.

UNIT - V

Cluster Analysis: Introduction, Types of Data in Cluster Analysis, A Categorization of Major Clustering Methods, Partitioning Method - k- Medoids Algorithm, CLARANS, Hierarchical Methods - BIRCH, ROCK Density-Based Methods - DBSCAN, Outlier Analysis.

Text books:

1. Jiawei Han & Micheline Kamber “Data Mining Concepts & Techniques”, Publisher Harcourt India. Private Limited, 2nd Edition.

Reference books:

1. Gupta G.K. “Introduction to Data Mining with case Studies”, PHI, New Delhi, 2006.
2. Berson A. & Smith S.J. “Data Warehousing Data Mining”, COLAP, TMH, New Delhi, 2004.
3. Dunham H.M. & Sridhar S. “Data Mining”, Pearson Education, New Delhi, 2006.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Third Semester

Branch	Subject Title	Subject Code
MCA	Data Communications and Computer Networks	MCA305

UNIT - I

Data Communications and Networking Overview: A Communications Model, Data Communications, Data Communication Networking. Protocol Architecture: The Need for a Protocol Architecture, A Simple Protocol Architecture, OSI, The TCP/IP Protocol Architecture

UNIT - II

Data Transmission: Concepts and Terminology, Analog and Digital Data Transmission, Transmission Impairments, Channel Capacity. Guided and Wireless Transmission: Guided Transmission Media, Wireless Transmission, Wireless Propagation, Line-of-Sight Transmission.

UNIT - III

Signal Encoding Techniques: Digital Data Digital Signals, Digital Data Analog Signals, Analog Data Digital Signals, Analog Data Analog Signals. Digital Data Communication Techniques: Asynchronous and Synchronous Transmission, Types of Errors, Error Detection, Error Correction, Line Configurations, Interfacing.

UNIT - IV

Data Link Control: Flow Control, Error Control, High-Level Data Link Control (HDLC). Multiplexing: Frequency Division Multiplexing, Synchronous Time Division Multiplexing, Statistical Time Division Multiplexing. Circuit Switching and Packet Switching: Switching Networks, Circuit-Switching Networks, Circuit-Switching Concepts, Control Signaling, Softswitch Architecture, Packet-Switching Principles, X.25, Frame Relay.

UNIT - V

Asynchronous Transfer Model: Protocol Architecture, ATM Logical Connections, ATM Cells, Transmission of ATM Cells, ATM Service Categories, ATM Adaptation Layer. Routing in Switched Networks: Routing in Circuit-Switching Networks, Routing in PacketSwitching Networks, Least-Cost Algorithms

Text Book:

1. Stallings W. "Data and Computer Communications", 7thEdition., Pearson Education./ PHI, New Delhi, 2006.

Reference Books:

1. Forouzan B. A., "Data Communications and Networking", 4th Edition. TMH, New Delhi, 2006. 2. Gupta P.C. "Data Communications and Computer Networks", PHI, New Delhi 2006.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Third Semester

Branch	Subject Title	Subject Code
MCA	Software Engineering Lab	MCA351

List of Programs as Assignments:

1. Draw use case diagram for online banking system.
2. Draw use case diagram for online library system
3. Draw use case diagram for online railway reservation system
4. Draw use case diagram for employee information system.
5. Draw use case diagram for inventory control system
6. Draw use case diagram for student information system.
7. Draw use case diagram for online hotel management system.
8. Draw use case diagram for online bus reservation system.
9. Draw use case diagram for online course registration system
10. Draw use case diagram for online teacher information system.
11. Draw sequence diagram for online banking system.
12. Draw sequence diagram for online library system.
13. Draw sequence diagram for online railway reservation system.
14. Draw sequence diagram for employee information system.
15. Draw sequence diagram for inventory control system.
16. Draw sequence diagram for student information system.
17. Draw sequence diagram for online hotel management system.
18. Draw sequence diagram for online bus reservation system.
19. Draw sequence diagram for online course registration system.
20. Draw sequence diagram for online teacher information system.
21. Draw activity diagram for online banking system.
22. Draw activity diagram for online library system.
23. Draw activity diagram for online railway reservation system.
24. Draw activity diagram for employee information system.
25. Draw activity diagram for inventory control system.
26. Draw activity diagram for student information system.
27. Draw activity diagram for online hotel management system.
28. Draw activity diagram for online bus reservation system.
29. Draw activity diagram for online course registration system
30. Draw activity diagram for online teacher information system.
31. Draw class diagram for online banking system.
32. Draw class diagram for online library system

33. Draw class diagram for online railway reservation system
34. Draw class diagram for employee information system.
35. Draw class diagram for inventory control system
36. Draw class diagram for student information system.
37. Draw class diagram for online hotel management system.
38. Draw class diagram for online bus reservation system.
39. Draw class diagram for online course registration system
40. Draw class diagram for online teacher information system.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Third Semester

Branch	Subject Title	Subject Code
MCA	Data Mining Lab	MCA354

- Q1. Build a Data Warehouse and Explore WEKA tool.
- Q2. Demonstration of preprocessing on various datasets.
- Q3. Demonstration of Association rule process on dataset using apriori algorithm.
- Q4. Demonstrate performance of classification on various data sets.
- Q5. Demonstrate performance of clustering on various data sets.
- Q6. Demonstrate performance of Regression on various data sets
- Q7. Implement following algorithms for various datasets A. Apriori Algorithm. B. FP-Growth Algorithm. C. K-means clustering.
- Q8. Implement Bayesian Classification for various datasets
- Q9 Implement Decision Tree for various datasets.
- Q10. Implement Support Vector Machines.
- Q11 Applications of classification for web mining.
- Q12. Case Study on Text Mining or any commercial application

Text Books :

1. Jiawei Han & Micheline Kamber - Data Mining Concepts & Techniques
Publisher Harcourt India. Private Limited.

Reference Books :

1. G.K. Gupta – Introduction to Data Mining with case Studies, PHI, New Delhi – 2006.

2. A. Berson & S.J. Smith – Data Warehousing Data Mining, COLAP, TMH, New Delhi – 2004.

3. H.M. Dunham & S. Sridhar – Data Mining, Pearson Education, New Delhi, 2006.

Computer Networks Lab



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Third Semester

Branch	Subject Title	Subject Code
MCA	Computer Networks Lab	MCA355

List of Programs as Assignments:

1. To familiarize with the Lab Network Topology, Locating different interfaces, routers and switches. Studying different pools of IP addresses.
2. Implement the data link layer framing methods such as character, character stuffing, and bit stuffing.
3. To learn and observe the usage of different networking commands e.g.PING, TRACEROUTE. Learning remote login using telnet session. Measuring typical average delays between different locations of the network.
4. What is the IP of the machine you are using? Compare it with the IP of your neighbors.Are the IPs of your neighbors same? Why or Why not?
5. Ping” is a tool used to determine if a server is responding and to estimate the round triptime of a message sent to that server. Use the ping command for the following URLs and record the success or failure statistics along with the average round trip time. a) google.com b) facebook.com c) bitmesra.ac.in
6. Trace the route that is taken when you try to access: a) google.com b) facebook.com c) rkdf.org.in
7. Network Commands on Linux / Unix
8. Implement on a data set of characters the three CRC polynomials – CRC 12, CRC 16 and CRC 32.
9. Implementation of Sub-netting and Super-netting.
10. To study different types of transmission media, various topologies, and configure modem of computer HUB and Switches.
11. Write a C/C++ program to determine if the IP address is in Class A, B, C, D, or E.
12. Write a C/C++ program to determine if the IP address is in Class A, B, or C.
13. Write a C/C++ program to translate dotted decimal IP address into 32 bit address.
14. To implement a routing protocol and check its connectivity in a variable length subnet masked network
15. Write a C/C++ program to perform bit stuffing and de-stuffing.
16. Implement Dijkstra’s algorithm to compute the Shortest path through a graph.
17. Take an example subnet graph with weights indicating delay between nodes. Now obtain Routing table at each node using distance vector routing algorithm
18. Take an example subnet of hosts. Obtain broadcast tree for it. 6.
19. Build implementations of the Internet protocols

20. Implementation of Stop and Wait Protocol and Sliding Window Protocol.
21. Write a code simulating ARP /RARP protocols.
22. Create a socket for HTTP for web page upload and download
23. Write a code simulating PING and TRACEROUTE commands.
24. Study and implement model for Socket Programming and Client – Server model.
25. Experiments with NS2(or any other simulator) to study behavior (especially performance of) link layer protocols such as Ethernet and 802.11 wireless LAN.
26. Experimental study of application protocols such as HTTP, FTP, SMTP, using network packet sniffers and analyzers such as Wireshark. Small exercise in socket programming in C/C++/Java.
27. Take a 64 bit playing text and encrypt the same using DES algorithm.
28. Write a program to break the above DES coding
29. Using RSA algorithm encrypts a text data and Decrypt the same
30. Applications using TCP and UDP Sockets like d. DNS e. SNMP f. File Transfer
31. Study of Network simulator (NS).and Simulation of Congestion Control Algorithms using NS
32. Echo client and echo server b. Chat c. File Transfer

Text books

1. William Stallings, Data and Computer Communication, Prentice Hall of India.
2. Behrouz A. Forouzan, Data Communication and Networking, McGraw-Hill.
3. Andrew S. Tanenbaum, Computer Networks, Prentice Hall.

Reference books

1. W. Richard Stevens, TCP/IP Illustrated, Volume 1, Addison-Wesley.
2. Douglas Comer, Internetworking with TCP/IP, Volume 1, Prentice Hall of India.



RKDF UNIVERSITY, RANCHI

New Scheme of Examination as per AICTE Flexible Curricula

MCA(Master in Computer Application)

SCHEME

Fourth Semester

S.No.	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
			Max	Max	Min	Max	Min
1	MCA401	Compiler Design	30	70	21	100	35
2	MCA402	System Simulation and Modeling	30	70	21	100	35
3	MCA403	Soft Computing	30	70	21	100	35
4	MCA404	Network Security and Cryptography	30	70	21	100	35
Practical							
			Total				
			Max		Min		
1	MCA451	Compiler Design Lab	50		25		
2	MCA452	System Simulation and Modeling Lab	50		25		
3	MCA453	Soft Computing	50		25		
4	MCA455	Project	200		70		



RKDF UNIVERSITY, RANCHI

New Scheme of Examination as per AICTE Flexible Curricula

MCA(Master in Computer Application)

Fourth Semester

Choice Based Credit System

MCA Year 1 Semester I						
S.No.	Subject Code	SUBJECT NAME	PERIODS			Credit
			L	T	P	
1	MCA401	Compiler Design	3	1	0	4
2	MCA402	System Simulation and Modeling	3	1	0	4
3	MCA403	Soft Computing	3	1	0	4
4	MCA404	Network Security and Cryptography	3	1	0	4
Practicals						
1	MCA451	Compiler Design Lab			2	1
2	MCA452	System Simulation and Modeling Lab			2	1
3	MCA453	Soft Computing Lab			2	1
4	MCA455	Project			2	4
Total					8	23



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Fourth Semester

Branch	Subject Title	Subject Code
MCA	Compiler Design	MCA401

UNIT - I

Introduction to Compilers and its Cousins, Structure of a Compiler, Science of building Compiler and its Application, Lexical Analyzer, Input Buffering, Specification and Recognition of Tokens, Introduction to Lex.

UNIT - II

Introduction to Syntax Analysis, Elimination of Ambiguity, Left Recursion and Left Factoring, Recursive and Non-Recursive Top-Down Parsers, Bottom-up Parsers: Shift Reduce Parser techniques and conflicts, all variants of LR Parsers, Handling Ambiguous grammar in Bottom-Up Parsing, Error handling while parsing, The Parser generator YACC.

UNIT - III

Syntax-Directed Definition(SDD), Evaluation Order of SDD's and its application, SyntaxDirected Translation Schemes and their Implementation.

UNIT - IV

Intermediate code Generation: Variants of Syntax Tree, Three Address Code, Translation of Expressions, Control flow, Back Patching , Run Time Environment: Storage Organization.

UNIT - V

Code Generation: Issues in its Design, Target Language, Addresses in Target Code, Basic Blocks and Flow Graphs, Optimization of Basic BlocksMachine Independent Optimization: Sources of Optimization, Data Flow analysis.

Text book:

1. Aho A. V., Lam M. S., Sethi R., Ullman J. D., Compilers, Principles, Techniques, and Tool, 2nd Edition, Pearson Education Asia.

Reference books:

1. Fischer C. N., LeBlanc R. J., Crafting a Compiler with C, Pearson Education Asia.
2. Loudon K. C., Compiler Construction, Principles and Practice, Thomson, Brooks/Cole.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Fourth Semester

Branch	Subject Title	Subject Code
MCA	System Simulation and Modeling	MCA402

UNIT - I

The concepts of a system, System Environment, Stochastic Activities, continuous and discrete systems, System Modeling, Types of models. System Studies: Subsystem, A Corporate Model, Environment segment, Production Segment, Management Segment, full Corporate Model, Types of System study, System Analysis, System Design, System Postulation.

UNIT - II

The technique of simulation, the Monte Carlo method, comparison of simulation and analytical methods, experimental nature of simulation, types of system simulation, numerical computation technique for continuous & discrete models, distributed lag models, cobweb models. Continuous system models, differential equations, analog computers & methods, hybrid computers, CSSLs, CSMP-III, Feedback Systems, Simulation of an Autopilot.

UNIT - III

Exponential Growth & decay models, modified exponential growth models, logistic curves, generalization of growth models, system dynamics diagrams, Simple system dynamics diagrams, multi-segment models, representation of time delays.

UNIT - IV

Evaluation of continuous probability functions, continuous uniformly distributed random numbers, a uniform random number numbers, generating discrete distributions, non-uniform continuously distributed random numbers, the rejection method. Random numbers Generators: Techniques for generating random numbers. Test for random numbers. Random vitiate Generation: Inverse transform technique, exponential distribution, uniform distribution.

UNIT - V

Queuing disciplines, measures of queues. Discrete events, representation of time, generation of arrival patterns, simulation of a telephone system, delayed calls, Simulation programming tasks, measuring utilization and occupancy.

Text books:

1. Gordon Geoffrey, "System Simulation", 2nd Edition, Pearson Education, 2007.
2. Banks J., Carson J. S., Nelson B.L., Nicol D.M. Nicol, "Discrete-Event System Simulation", 4th Edition, Pearson Education, 2007.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Fourth Semester

Branch	Subject Title	Subject Code
MCA	Soft Computing	MCA403

UNIT - I

Introduction to Artificial Intelligence System, Neural Network, Fuzzy Logic & Genetic Algorithm. Fuzzy Set Theory: Fuzzy Versus Crisp, Crisp Set, Fuzzy Set, Crisp Relation, Fuzzy Relations.

UNIT - II

Fuzzy System: Crisp Logic, Predicate Logic, Fuzzy Logic, Fuzzy Rule Based System, Defuzzification Methods, and Applications.

UNIT - III

Genetic Algorithms, Basic Concepts, Creation Of Offspring, Working Principle, Encoding, Fitness Function, Reproduction. Genetic Modeling, Inheritance Operations, Cross Over, Inversion And Deletion, Mutation Operator, Bit Wise Operators, Generation Cycle, Convergence Of Genetic Algorithm, Application, Multi-Level Optimization, Real Life Problems, Difference And Similarities Between GA And Other Traditional Methods, Advanced In GA.

UNIT - IV

Fundamentals Of Neural Networks, Basic Concepts Of Neural Network, Human Brain, Model Of An Artificial Neuron, Neural Network Architectures, Characteristic Of Neural Networks, Learning Method, Taxonomy Of Neural Network Architectures, History Of Neural Network Research, Early Neural Network Architectures, Some Application Domains.

UNIT - V

Back Propagation Network Architecture Of Back Propagation Network, Back Propagation Learning, Illustration, Applications, Effect Of Tuning Parameters Of The

Back Propagation Neural Network, Selection Of Various Parameters In BPN, Variations Of Standard Back Propagation Algorithm. Associative Memory And Adaptive Resonance Theory, Autocorrelations, Hetrocorrelators , Multiple Training Encoding Strategy, Exponential BAM, Associative Memory For Real Coded Pattern Pairs, Applications, Introduction To Adaptive Resonance Theory, ARTI, Character Recognition Using ARTI

Text Book:

1. Rajasekharan S. &Vijayalakshmi G. A. “Neural Network Fuzzy Logic and GenticAlgorithm Synthesis and Applications”, Prentice Hall of India PLT, Pai, 2004.

Reference Book:

2. 1. Jang JyhShing R, Sun C. T., Mizutani E. “Neuro Fuzzy and Soft Computing –A Computational Approach to Learning and Machine Intelligence”, Prentice Hall of India, 1997.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Fourth Semester

Branch	Subject Title	Subject Code
MCA	Network Security and Cryptography	MCA404

UNIT - I

Foundations – Protocol Building Blocks - Basic Protocols - Intermediate Protocols - Advanced Protocols - Zero-Knowledge Proofs - Zero-Knowledge Proofs of Identity - Blind Signatures - Identity-Based Public-Key Cryptography.

UNIT - II

Key Length - Key Management – Public Key Cryptography versus Symmetric Cryptography - Encrypting Communications Channels - Encrypting Data for Storage - Hardware Encryption versus Software Encryption - Compression, Encoding, and Encryption - Detecting Encryption – Hiding and Destroying Information.

UNIT - III

Information Theory - Complexity Theory - Number Theory - Factoring - Prime Number Generation - Discrete Logarithms in a Finite Field - Data Encryption Standard (DES) – Lucifer - Madryga - NewDES - GOST – 3 Way – Crab – RC5 - Double Encryption - Triple Encryption - CDMF Key Shortening - Whitening.

UNIT - IV

Pseudo-Random-Sequence Generators and Stream Ciphers – RC4 - SEAL - Feedback with Carry Shift Registers - Stream Ciphers Using FCSRs - Nonlinear-Feedback Shift Registers - SystemTheoretic Approach to Stream-Cipher Design - Complexity-Theoretic Approach to StreamCipher Design - N- Hash - MD4 - MD5 - MD2 - Secure Hash Algorithm (SHA) - OneWay Hash Functions Using Symmetric Block Algorithms - Using Public-Key Algorithms - Message Authentication Codes

UNIT - V

RSA - Pohlig-Hellman - McEliece - Elliptic Curve Cryptosystems -Digital Signature Algorithm (DSA) - Gost Digital Signature Algorithm - Discrete Logarithm Signature Schemes - OngchnorrShamir -Cellular Automata - Feige-Fiat-Shamir -Guillou-Quisquater - Diffie-Hellman - Station- to-Station Protocol -Shamir's Three-Pass Protocol - IBM Secret-Key Management Protocol - MITRENET - Kerberos - IBM Common Cryptographic Architecture.

Text Books:

1. Schneier Bruce, "Applied Cryptography: Protocols, Algorithms, and Source Code in C", 2 nd Edition, John Wiley & Sons, Inc, 1996.
2. Mao Wenbo, "Modern Cryptography Theory and Practice", Pearson Education, 2004.
3. KahateAtul, "Cryptography and Network Security", Tata McGrew Hill, 2003.

Reference Book:

1. Stallings William, "Cryptography & Network Security Principles and Practice", Pearson Education.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Fourth Semester

Branch	Subject Title	Subject Code
MCA	Compiler Design Lab	MCA451

1. C program to count white spaces, numbers, words in a file
2. C program to design Finite automata to identify different tokens(identifiers, constants, operators, etc.).
3. Count number of a's in given string.
4. Identify different patterns like aa, ab, not containing a, etc. in given string .
5. Lex program to Identify all tokens of C programs.
6. .Design and Code individual programming code with all possible tokens in programming language
7. Starting and ending with 'a'.
8. # a's divisible by 2 or b's divisible by 3.
9. 4th Symbol 'a' from RHS.
10. Output code after removing white spaces and comment.
11. Build parsers using yacc for $L(G)=\{a^n b^n \mid n \geq 1\}$ over $\{a,b\}$
12. Build Parser using yacc for $L(G)$ where rule set of G is $\{ S \rightarrow aSb, S \rightarrow bSa, S \rightarrow c \}$ over $\{a,b,c\}$
13. Build parser using yacc to convert the infix expression to postfix expression.
14. Build a calculator in yacc which takes expression in postfix notation.
15. Build parsers using yacc to convert the prefix expression into the postfix expression.
16. Build parsers using yacc to validate the C statements. E.g int a,b,c;(valid)
17. Build calculator in yacc.

Text books

1. lex&yacc (2nd ed.) :O'Reilly & Associates, Inc. Sebastopol, CA, USA ©1992 .

Reference books

1. Lex &Yacc:O'Reilly & Associates, Inc. Sebastopol, CA, USA ©1992.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Fourth Semester

Branch	Subject Title	Subject Code
MCA	System Simulation and Modeling Lab	MCA452

1. Introduction to 3D Studio Max.
2. Exploring the Max Interface. Lab Assignment
3. Understanding 2D Splines & Shape
4. Convert 2D to 3D object using extrude, bevel, loft, terrain etc.
5. Modeling with polygon objects
6. Building Simple & Complex Scene
7. Creating key frames & Auto Key/Set Key
8. Animating with simple controllers.
9. Simulation & Effects.
10. Bind to space warp objects.
11. Configuring & Aiming Cameras
12. Using Camera Motion Blur & Depth of Field.
13. Texturing with Max.
14. Create & Apply standard material
15. Rendering with V-Ray.
16. Introduction to Scene
17. Create & Assign Textures.
18. Basic Settings for Texturing.

TEXT BOOKS

1. Beginning Blender: Open Source 3D Modeling, Animation, and Game Design By Lance Flavel
2. 3d'sMax5Fundamentals nside3dsmax7 By Ted Boardman

REFERENCE BOOKS

1. 3D Modeling and Animation By Michael G.



R.K.D.F. UNIVERSITY, RANCHI

MCA(Master in Computer Application)

Fourth Semester

Branch	Subject Title	Subject Code
MCA	Soft Computing Lab	MCA453

1. Implementation of a 2 input AND and OR logic function using perceptron. Start with different set of initial weights and show that there are more than one solution to the problem.
2. Develop an algorithm using Hebbian Learning rule to solve NAND and NOR problem. Assume the elements of initial weight matrix as random value and study the effect of different learning rates.
3. Solve the following Boolean function using Perceptron learning rule:
$$Y = x_1' x_2' x_3' + x_1' x_2' x_3 + x_1 x_2' x_3' + x_1 x_2 x_3$$
4. Consider the following function:
$$F(x) = x_1^3 + 2x_1x_2 - x_1^2x_2^2$$

Perform two iterations of Newton's method from the initial guess $x_0 = [1 \ -1]^T$.
5. Demonstrate that EX-OR and XNOR gate is a non-linearly separable problem. Design a MLP for the purpose and train it using BP algorithm. Assume the use of a logistic function for the nonlinearity.
6. Given the dataset $\{-2, -1, 0, 1, 2\}$ with targets $t(-2)=0$, $t(-1)=0.25$, $t(0)=0.5$, $t(1)=0.75$ and $t(2)=1$. Determine the weights of all neurons with sigmoid transfer function such that the MSE is almost zero and if we use the neural network with five neurons in the first layer and one neuron in the second layer.
7. Using a MLP with BP algorithm approximate the function $y=1+\sin(2\pi x)$ for $-1 \leq x \leq 1$.

Draw the actual output and simulated output. Show the signal matching after 100/ 500/ 1000 iterations.

8. Implement a 2-input EX-OR gate using regularized RBF (use 4 centres). Plot the output surface for different input variables. Extend this for a 3 input RBF using 8 centers. Tabulate the input output pattern mapping for each case.

9. Using Mamdani and Sugeno model Fuzzy system, design a non-stationary time series prediction network to predict the output against time sample.

10. Minor Project

REFERENCE BOOKS

1. S. Haykin, *Neural Networks - A Comprehensive Foundation*, Pearson Education, India
2. Jang, Sun and Mizutani, *Neuro-Fuzzy and Soft-Computing – A computational approach to learning and machine intelligence*, Prentice Hall of India